



Introduction to natural preservatives for food

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1.1 Introduction

Artificial preservatives are used in food industry for the shelf life extension of different food commodities. However, there is a growing concern among consumers about the usage of synthetic preservatives, which have been limited due to their negative effects on human health. Furthermore, consumers are asking for food products, which are free from any artificial additives and contain ingredients/additives, which are natural and perceived to be good for health. In order to satisfy consumer demand, the food industry is searching for natural alternatives that exhibit strong antimicrobial, antioxidant, and antibrowning properties and can be used for shelf life enhancement of different food commodities.

Natural preservatives are chemical agents that are derived from plant, animal, or microbial sources for shelf life enhancement of food products (Hugo & Hugo, 2015). Natural preservatives ensure food safety without impairing the negative effects on human health. Several natural compounds have been screened and investigated for utilization as natural preservatives in the food industry. The research includes the study of antibacterial and antioxidant properties of compounds originating from plant, microbial, and animal sources as natural preservatives.

In recent years, natural preservatives are gaining importance as they do not have any negative health on human health. The main aim of using natural preservatives in the food industry is to prevent the growth of undesirable microorganisms. Plant-based preservatives (green tea, rosemary extract, etc.), animal-based (chitosan from crustacean shells),

and microbial-based (bacteriocins) have been shown antimicrobial and antioxidant properties (Paglarini et al., 2023). Natural preservatives are applied to the food commodities in product formulation, coated on the surface of foods, or incorporated into the packaging material. Natural preservatives have found application in the food industry for a wide range of products such as meat, poultry, fruits, vegetables, dairy, etc. with promising results.



1.2 Antimicrobials

Antimicrobial agents are substances that inhibit the growth of microorganisms and are used to preserve the food. The utilization of preservatives derived from natural sources has a high potential against food spoilage microorganisms. Different types of plant, animal, and microbial origin sources can be used as antimicrobials for food applications (Baptista et al., 2020).

Plants have been used as a source of natural antimicrobial agents due to the presence of bioactive compounds such as phenolics, alkaloids, and terpenes. Plants have diverse secondary metabolic compounds with varied antimicrobial activity. The phenolic and polyphenolic groups are among the major groups found in the secondary metabolite molecule. A few subgroup compounds, including flavonoids, flavones, flavonols, coumarins, quinones, phenolic acids, tannins, and phenols are also significant for inhibiting microbial activity. The chemical with a hydroxyl (-OH) group is called phenol. It is believed that the locations and number of phenol groups present in the compound correspond to how hazardous they are to microorganisms. The toxicity increases with the number of hydroxyl groups present. These compounds are usually present in different parts of plants such as leaves, roots, and bark (Bhardwaj & Dubey, 2021).

The antimicrobial mechanism of natural preservatives is related to the phospholipid bilayer of the cell membrane, which causes the rupture of cell membranes and leads to the compromising of the genetic material of microbes. The process leads to the formation of hydroperoxidase of fatty acids by the oxygenation of unsaturated fatty acid and coagulation of cytoplasm, which results in damage of lipids and proteins and distortion of the proton motive force, electron flow, and/or active transport. They also result in the inhibition of protective enzyme activity and consequently block one or more biochemical pathways (Viuda-Martos et al., 2011).

Essential oil is also one of the phytochemicals used for food preservation. Essential oils are volatile substances that have a sweet scent and oily viscosity. These essential oils are extracted from different parts of plant such as flowers, leaves, and seeds. The antimicrobial activity of essential oils is attributed to the oxygenated terpenoids including phenolic terpenes and alcohol. Essential oils from plant sources are able to reach inside the microbial in liquid and gas phases. The essential oils will react with lipids at the cell membrane and will degrade the membrane of microbes. This will lead to the increase of permeability of the cell and damage the cytoplasmic membrane. This will cause the leakage of cell contents and the coagulation of cytoplasmic contents and destroy the native structure of the cell.

The hydrophobic nature of essential oils' constituents enables them to act on the lipids found in bacterial and mitochondrial cell membranes, causing structural distortion and increasing the susceptibility of the microorganisms to the antimicrobial action that releases cellular content. Therefore, the antibacterial activity and effect of essential oils can be greatly influenced by the chemical structure of their constituents of essential oils.

Animals also have been used as sources of antimicrobial agents for food applications. The lysozyme enzyme found in eggs is frequently used as a preservative and antibacterial for fruits, meat, poultry products, and other foods. Lysozyme has an antimicrobial capacity due to its ability to hydrolyze β -1,4 linkage between N-acetylglucosamine and N-acetylmuramic acid at the microbial cell wall. Lysozyme is used commercially for control of late blowing in semihard cheese that is caused by *Clostridium tyrobutyrium*. Lysozyme is generally used against gram-positive bacteria. Chitosan is the animal-derived antimicrobial used in different food commodities against a range of foodborne commodities. Chitosan is more effective against gram-negative as compared to gram-positive bacteria. The preservative property of chitosan is due to its ability to cause permeability of the cell membranes and inhibition of various enzymes. The inhibition of bacterial growth is believed to be related to chitosan ability to establish electrostatic interactions between chitosan cationic amino groups and anions, such as N-acetylmuramic, neuraminic, and sialic acid present in the cell wall. Moreover, hydrophobic interactions resulting from the presence of N-acetylated residues in chitosan also contribute to bacteriostatic properties of chitosan.

Bacteria produce a variety of compounds that are active against other bacteria. These active bacteria can hinder or stop the growth of spoilage-causing microbes. The substance that can function as an antibacterial agent against spoiling or disease microbes is a protein complex called bacteriocins. Both gram-positive and gram-negative bacteria have the ability to

produce bacteriocins. These proteinaceous substances permeate the cytoplasmic membrane and cause intracellular metabolites leakage, which results in membrane depletion. In addition to bacteriocins, there are more active bacteria such as reuterin and pediosin that inhibit the growth of spoilage microorganisms. Lactic acid bacteria, such *Lactobacillus acidophilus*, produce bacteriocins as a result of their metabolic activities. Bacteriocins have the potential to inhibit the growth of foodborne pathogens such as *Clostridium botulinum*, *Enterococcus faecalis*, *Listeria monocytogenes*, etc. The bacteriocins are safe to use as natural antimicrobials as they can be degraded by protease. Nisin is produced from *Lactococcus lactis* and is capable of inhibiting the gram-positive bacteria.

Organic acids are naturally found in several foods including malic acid (apple), ascorbic acid (orange), tartaric acid (grapes), citric acid (lemon), and acetic acid (vinegar). Organic acids are widely employed as food preservatives and are generally recognized as safe (GRAS). The undissociated molecule or ester of organic acids is responsible for potential antimicrobial activity based on the lowering of the pH of a food commodity. The undissociated molecules are hydrophobic and are able to cross the lipid membrane of microorganisms and enter into their cytoplasm, thus resulting in cell death. Inside the cytoplasm of the microbial cell, the organic acid molecules encounter a higher pH, facilitating the dissociation of acid into charged anions and protons that could not cross the plasma membrane. The accumulating H^+ ions will then decrease intracellular pH toward harsh physiological conditions inducing cellular damage and modification of the functionality of enzymes, structural proteins, and DNA. Furthermore, the generated anions were toxic, which led to inhibiting metabolic reactions and causing cell death.

Furthermore, the release of these anions contributes to the increases of osmotic pressure in the cytoplasm that is deleterious to cytosolic enzymes. Weak organic acids such as sorbic and benzoic acids could act in their undissociated forms by penetrating microbial cells and inactivating them by lowering their internal pH level or by interfering with metabolic reactions. Therefore, the organic acids were assigned to exhibit stronger antimicrobial effects than the highly dissociating inorganic acids at the same pH level due to their amount of undissociated molecules and their cell penetration capacity, suggesting their application in food biopreservation.

Organic acids such as citric and ascorbic acid act as antioxidants in different food commodities. The organic acids act as pro-oxidants and chelating the transition metals present in the medium and make them unavailable to act as oxidation propagators.



1.3 Antioxidants

Antioxidants are substances that are used in food commodities to extend the shelf life by inhibiting oxidation reactions. Currently, food industry provides a variety of natural antioxidants for application in food products, justifying the need, importance, and efficiency for delaying or inhibiting lipid oxidation reactions. Lipid oxidation results in a change in color, flavor, and texture, and also destroys the fat-soluble vitamins. Moreover, oxidative reactions damage biological membranes, protein, and enzymes, which may pose a threat to human health. These antioxidants are used for the control of lipid oxidation in different food industries such as nuts, meat, poultry, and fish ([Hadidi et al., 2022](#)).

Plants are a potent source of antioxidant agents due to the presence of various types of active compounds. Antioxidants are found in different parts of plants such as fruits, bark, leaves, and flowers. Phenolic compounds, tocopherols, and carotenoids are bioactive compounds from plant sources that are GRAS as food additives for human consumption. The antioxidant properties of plant extracts are primarily due to the presence of large amounts of phenolic compounds ([Ofosu et al., 2020](#)). Phenolic compounds contain one or more hydroxyl groups, which can provide the hydrogen atoms to free radicals thus inhibiting the oxidative reactions. For the last few decades, essential oils have been used as an antioxidant agent for food preservation due to the presence of different phenolic compounds. The polyphenol quality and quantity in plants significantly vary and depend on many factors such as plant genetics, variety, climatic conditions, soil composition, and maturity stage. The natural antioxidants were extracted from different plant sources such as grape, green tea, mate, pomegranate, and rosemary and used in deboned poultry meat ([Paglarini et al., 2023](#)).



1.4 Antibrowning agents

Enzymatic browning in fruits and vegetables occurs when enzymes, oxygen, and phenolic compounds come together for enzymatic reactions. In fresh fruits and vegetables, melanin compounds are not formed due to the compartmentalization of cells that keeps phenols and other substances separated from enzymes. Enzymatic browning is mainly caused by the

polyphenol oxidase activity that is naturally present in food that shortens the shelf life of fruits and vegetables. Enzymatic browning always takes place after harvest, during postharvest operations, and also during storage. The processing operations such as peeling, slicing, and dicing trigger the browning reactions. Therefore there is a need for suitable and acceptable inhibitors, which will prevent the browning and increase the food marketability. There is also a growing demand of consumers for food products that are safer and sustainable than synthetic ones.

Natural preservatives have been used for the control of browning reactions in food products (Navina et al., 2023). The sweet potato polyphenol oxidase activity was significantly inhibited by the extracts of honey, pineapple, onion, and chili pepper, which are safe from human consumption (Lim et al., 2019). The inhibition capacity of honey is due to the chelating efficiency, which involves peptides with a molecular weight of 600 and may chelate copper of active size polyphenol oxidase (PPO) and thus inactive the enzyme. Additionally, the high percentage of phenolics in honey also contributes to this inhibitory effect (Haard & Simpson, 2000). Natural antibrowning agents have potentially been used to control the browning of ginger. Various extracts have been utilized for the inhibition of ginger PPO and heated chili pepper extract was the best natural inhibitor with a PPO inhibition capacity of 47.97% (Lim & Wong, 2018). Ginger has been also used as an antibrowning agent against PPO activity in *Annona muricata* and *Musa acuminata* that shows its effectiveness in controlling browning (Weerawardana et al., 2020). Ginger consists of different constituents, including ascorbic acid, caffeic acid, beta-carotene, curcumin, beta-sitosterol, limonene, etc. The inhibition activity of ginger is mainly due the ascorbic acid. The antibrowning potential of different plant extracts was studied by Dias et al. (2020). The study demonstrated that apple pomace extract is effective for the control or delay of browning in fresh-cut pear. Furthermore, they reported that effectiveness is related to the enriched phenolic compounds. The antimicrobial agents from onion have been used to prevent the browning in food products (Lee et al., 2002).



1.5 Importance of natural preservatives

The food industry is using different types of synthetic preservatives to enhance the shelf life of food products. Synthetic preservatives

have many advantages such as they are cheap, readily available, and easy to use. Synthetic preservatives include benzoates, sulfites, nitrates, sorbates, butylated hydroxytoulene (BHT), butylated hydroxyanisole (BHA), etc., which possess many negative effects on human health. Their use has increased complications including carcinogenicity, liver, renal, respiratory, and neurological complexities, and other ailments. This health danger resulted in the need and demand for natural preservatives among consumers. Natural preservatives are safer alternatives to synthetic ones. Natural preservatives reduce the negative side effects on health while extending the shelf life of food. Therefore natural food preservatives should be preferred above artificial ones. Natural preservatives have been widely accepted by consumers who are looking for natural and healthy products. Consumers are nowadays more aware of the safety of any additives in food and keen to use natural preservatives, considered more natural and healthy.

Natural preservatives are mainly obtained from plant sources and microbial or animal sources. Essential oils derived from plants, bacteriocins from microbial sources, organic acids, and naturally occurring polymers (chitosan) have been used for different food applications as natural preservatives. Furthermore, these natural preservatives have many nutritional and health benefits due to the presence of different phytochemicals in addition to the preservation effect. The important vitamins including vitamin C and E having the preservative efficiency as act free radical scavengers in addition to health benefits.



1.6 Conclusion

Natural preservatives have shown the potential to replace chemical preservatives in the food industry for food preservation. Natural preservatives are suitable alternatives for artificial ones, which are safer, highly acceptable among consumers, and also lessen the burden of metabolic diseases. The preservative properties of natural ingredients are due to the presence of phenolic compounds including flavonoids and terpenes. Natural preservatives have been identified from different sources such as plant, animal, and microbial origin. These preservatives have found applications as antimicrobial, antioxidant, and antibrowning agents.

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